



1. Which of the following statements are correct?
- In a directed graph, the sum of lengths of all the adjacency list is $|E|$.
 - The adjacency matrix requires $\theta(V^2)$ memory.
 - In an undirected graph, the sum of lengths of all the adjacency list is $|E|$.
 - The memory requirement of adjacency list is dependent on the number of edges in the graph.

Choose the correct answer from the options given below:

- (1) A, B, C only (2) B, C, D only
(3) A, B, D only (4) A, C, D only

2. A tree used to represent arithmetic expressions is called _____ tree.

- (1) Threaded (2) Game
(3) Red-Black (4) Expression

3. Choose the odd one from the following:

- (1) Write (2) Read
(3) Knowledge (4) Learn

4. Match List I with List - II.

	List-I		List-II
A.	4, 7, 12, 19, 28, ?	I.	5
B.	28, 33, 31, 36, ?, 39	II.	39
C.	125, 80, 45, 20, ?	III.	50
D.	1, 3, 4, 8, 15, 27, ?	IV.	34

Options:

- (1) A-I, B-II, C-III, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-II, C-II, D-IV

5. The occurrence of an event is usually signaled by an interrupt from software by executing a special operation, which is called as _____:

- (1) System call
(2) Bootstrap Program
(3) Service Routine
(4) Interrupt - Vector

6. Match List - I with List - II.

	List - I		List - II
A.	Breadth first search	I.	Shortest paths
B.	Sparse Graph	II.	Adjacency list representation
C.	Collision Resolution	III.	Open addressing
D.	Linked list	IV.	Self referential structures

Options:

- (1) A-I, B-II, C-III, D-IV (2) A-I, B-II, C-IV, D-III
(3) A-II, B-I, C-III, D-IV (4) A-IV, B-III, C-II, D-I

7. Cooperating process require an Interprocess communication mechanism. This works on which of the following fundamental models?

- A. Shared Memory B. Cascading Termination
C. Message Passing D. Cooperating

Choose the correct answer from the options given below:

- (1) C and D only (2) A and C only
(3) A and D only (4) B and D only

8. Match List I with List - II (Scheduling Description).

	List-I (Scheduling)		List-II (Description)
A.	First-Come, First Served	I.	CPU is allocated to the process with the highest priority
B.	Shortest-Job-First	II.	The Process that requests the CPU first is allocated first
C.	Priority Scheduling	III.	Algorithm associates with each process the length of next CPU burst
D.	Round Robin	IV.	A pre-emption is added to switch between processes



Options:

- (1) A-I, B-II, C-IV, D-III (2) A-II, B-III, C-IV, D-I
(3) A-II, B-III, C-I, D-IV (4) A-III, B-I, C-II, D-IV

9. Consider a procedure LIST-SEARCH (L, K) finds the first elements with key K in list L by simple linear search, returning a pointer to its element. Arrange the following according to correct order of the execution.

- A. $x = L \rightarrow \text{head}$
B. While $x \neq \text{NIL}$ and $x.\text{key} \neq K$
C. return x
D. $x = x.\text{next}$

Options:

- (1) A, B, D, C (2) B, C, D, A
(3) A, D, B, C (4) A, B, C, D

10. Arrange the following steps in sequence to find the variance of individual observations.

- A. Compute the mean $\bar{X}$ of the given observations $x_1, x_2, \dots, x_n$.
B. Square the deviations obtained and obtain the sum $\sum_{i=1}^n (x_i - \bar{X})^2$
C. Divide the sum $\sum_{i=1}^n (x_i - \bar{X})^2$ by n.
D. Take the deviations of the observations from mean i.e. $x_i - \bar{X}, i = 1, 2, \dots, n$.

Choose the correct answer from the options given below:

Options:

- (1) A, B, C, D (2) A, D, B, C
(3) A, C, B, D (4) B, C, D, A

11. A half adder has:

- (1) One input and two outputs
(2) Two inputs and two outputs
(3) Three inputs and one output
(4) Two inputs and one output

12. _____ is a memory management scheme that permits the physical address space of a process to be non-contiguous.

- (1) Dynamic loading
(2) Address binding
(3) Paging
(4) Partitioning

13. Choose the correct alternative that will continue the same pattern:

240, _____, 120, 40, 10, 2

- (1) 180 (2) 240
(3) 420 (4) 480

14. If a matrix is both symmetric and skew symmetric, then:

- (1) A is a diagonal matrix
(2) A is a zero matrix
(3) A is a scalar matrix
(4) A is a square matrix

15. **Assertion (A):**

Copper is used to make electrical wires.

Reason (R):

Copper has a very low electrical resistance.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct

16. Which of the following requirement must be satisfied to solve a critical section problem?

- A. Mutual Exclusion B. Race Condition
C. Progress D. Bounded waiting

Options:

- (1) A, B and D only (2) A, C and D only
(3) B, C and D only (4) A, B and C only

17. **Assertion (A):**

The sequence $\frac{1}{3}, -\frac{1}{2}, \frac{3}{4}, -\frac{9}{8}, \dots$ is an AP.

Reason (R):

The constant sequence is the only sequence which is both an A.P. as well as G.P.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct



18. In a university, out of 100 students 15 offered Mathematics only; 12 offered statistics only; 8 offered Physics only; 40 offered Physics and Mathematics; 20 offered Physics and Statistics; 10 offered Mathematics and Statistics; 65 offered Physics. What is the number of students who did not offer any of the three subjects?

- (1) 1 (2) 2
(3) 3 (4) 4

19. Arrange the following games in decreasing order, on the basis of the players in their playing team.

- A. Basketball
B. Table Tennis
C. Lawn Tennis Double
D. Cricket

Options:

- (1) B, C, A, D (2) A, B, C, D
(3) D, A, B, C (4) D, A, C, B

20. The complement of $(\bar{A} + \bar{B})$ is:

- (1) $\bar{A} + \bar{B}$ (2) $\bar{A} \cdot \bar{B}$
(3) $A + B$ (4) $A \cdot B$

21. Match List I with List - II.

	List-I		List-II
A.	4:17 :: 7:?	I.	70
B.	720:840 :: 60:?	II.	82
C.	225:256 :: 289:?	III.	324
D.	21:3 :: 574:?	IV.	50

Choose the correct answer from the options given below:

- (1) A-IV, B-I, C-III, D-II (2) A-I, B-II, C-III, D-IV
(3) A-I, B-IV, C-II, D-III (4) A-IV, B-III, C-II, D-I

22. Arrange the following time complexities in increasing order:

- A. Linear search worst case
B. Quick sort expected running time
C. Binary search best case
D. Bubble sort worst case

Choose the correct answer from the options given below:

- (1) A, C, B, D (2) C, D, B, A
(3) C, A, B, D (4) A, B, C, D

23. In the questions given below, two statements are given following by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. Read the conclusion and then decide which of the given conclusions logically follows from the two given statement, disregarding commonly known facts:

Statements:

All windows are doors. No door is wall.

Conclusions:

- I. No window is wall II. No wall is door
Choose the correct answer from the options given below:

- (1) Only conclusion I follows.
(2) Only conclusion II follows.
(3) Neither conclusion I nor II follows.
(4) Both conclusion I and II follow.

24. Arrange the following steps of K-Map simplification in the correct order:

- A. Write the minimized SOP/POS expression
B. Plot the truth table outputs (1_s and 0_s) on the K-Map
C. Identify the prime implicants and essential prime implicants
D. Draw groups (loops) of adjacent 1_s (or 0_s) that are powers of 2

Choose the **correct** answer from the options given below:

- (1) A, B, C, D (2) B, C, A, D
(3) C, A, B, D (4) B, D, C, A

25. Which of the following statements are false about two circles whose equations are given below?

$$x^2 + y^2 - 10x - 10y + 41 = 0 \text{ and } x^3 + y^2 - 22x - 10y + 137 = 0$$

- A. Circles have same centre
B. Circles have no meeting point
C. Circles have only one meeting point
D. Circles have two meeting points

Choose the correct answer from the options given below:

- (1) A, B, C only (2) B, C, D only
(3) A, B, D only (4) A, C, D only



26. A JK flip flop behaves as a T flip flop when:
- (1) $J = 0, K = 1$ (2) $J = 1, K = 0$
 (3) $J = 1, K = 1$ (4) $J = 0, K = 0$
27. Arrange the following semaphore operations in the correct sequence for the producer process in the Bounded-Buffer Problem using semaphores empty, full, and mutex:
- A. Signal $\rightarrow$ indicating that the new item (full) has been produced and added to the buffer
 B. Wait $\rightarrow$ Checks if there is space in the (empty) buffer to add new item
 C. Produce an item and store it in the buffer
 D. Wait $\rightarrow$ Centres the critical section to (mutex) insert the item safely
 E. Signal $\rightarrow$ Leaves the critical section after (mutex) inserting the item

Choose the correct answer from the options given below:

- (1) A, E, C, B, D (2) C, B, A, D, E
 (3) D, E, A, C, B (4) B, D, C, E, A

28. Match List I with List - II. List - I

A.	Decoder	I.	Selects one input out of many
B.	Multiplexer	II.	Converts binary input into 2^{nd} outputs
C.	Encoder	III.	Converts 2^{nd} inputs into binary code
D.	Demultiplexer	IV.	Routes one input to several outputs

Choose the correct answer from the options given below:

- (1) A-I, B-II, C-III, D-IV (2) A-II, B-I, C-III, D-IV
 (3) A-III, B-II, C-IV, D-I (4) A-IV, B-III, C-II, D-I

29. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The binary search tree property allows to print out all the keys in a binary search tree in a sorted order by a simple recursive algorithm called an inorder tree walk.

Reason (R): The inorder tree walk algorithm prints the key of the root of subtree between printing the values in its left subtree and printing those in its right subtree.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (3) (A) is correct but (R) is not correct
 (4) (A) is not correct but (R) is correct

30. Match List - I with List - II.

	List - I		List - II
A.	Taxonomy	I.	Biography
B.	Pedology	II.	History
C.	Man	III.	Soils
D.	Nation	IV.	Classification

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-II, D-I (2) A-I, B-II, C-III, D-IV
 (3) A-II, B-III, C-I, D-IV (4) A-IV, B-III, C-I, D-II

31. Let S and T be symmetric matrices of the same order, which of the following statements are true ?

- A. $S + T$ is a symmetric matrix
 B. $ST - TS$ is a skew symmetric matrix
 C. $ST + TS$ is a symmetric matrix
 D. $ST - TS$ is a symmetric matrix

Choose the correct answer from the options given below:

- (1) A, B, C only (2) A, B, D only
 (3) B, C, D only (4) A, C, D only

32. Given below are two statements :

one is labelled as Assertion (A) and the other is labelled as Reason (R).

Let A and B be two symmetric matrices of order 3.

Assertion (A): A (BA) and (AB)A are symmetric matrices.

Reason (R): AB is symmetric matrix, if matrix multiplication of A with B is commutative.

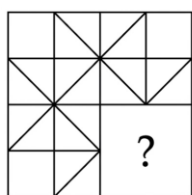




In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct

33. Select a figure from amongst the four alternatives which when placed in the blank space of fig (?) would complete the pattern.



34. Which of the following algorithms have same order of complexity ?

- A. Deletion from Queue
B. Worst case Search in Binary Search tree
C. Insertion in Stack
D. Quick Sort worst Case

Choose the correct answer from the options given below:

- (1) C and D only (2) A and C only
(3) A and B only (4) B and C only

35. Which of the following statement are false about

$$\lim_{y \rightarrow 0} \frac{\sqrt{1+\sqrt{1+y^4}} - \sqrt{2}}{y^4}$$

- A. Limit exists and equals $\frac{1}{4\sqrt{2}}$
B. Limit does not exist
C. Limit exists and equals $\frac{1}{2\sqrt{2}}$
D. Limits exists and equal $\frac{1}{2\sqrt{2}(\sqrt{2}+1)}$

Choose the correct answer from the options given below:

- (1) A, B, C only (2) A, B, D only
(3) B, C, D only (4) A, C, D only

36. Match List - I with List - II.

	List - I		List - II
A.	Minimum Spanning Tree	I.	Height balanced
B.	Topological Sort	II.	Stack
C.	Recursion	III.	Kruskal's Algorithm
D.	AVL Trees	IV.	DFS

Choose the correct answer from the options given below:

- (1) A-I, B-II, C-III, D-IV (2) A-IV, B-III, C-II, D-I
(3) A-II, B-I, C-III, D-IV (4) A-III, B-IV, C-II, D-I

37. Cub is related to Tiger in the same way as Fawn is related to _____.

- (1) Monkey (2) Stag
(3) Ass (4) Sheep

38. If the curves $x = y^2$ and $xy = k$ cut at right angles. Then the value of k is:

- (1) $\frac{1}{2\sqrt{2}}$ (2) 2
(3) 4 (4) $\frac{1}{8}$

39. Given below are two statements :

one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Leaves of Plants are green.

Reason (R) : Plants contain Chromoplast, the green pigment

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct



40. In a certain code, CONDEMN is written as CNODMEN. TEACHER will be written as (T _ _ C E _ _) Fill in the blanks by identifying the correct sequence.

A. H B. A C. E D. R

Choose the correct answer from the options given below:

(1) A, B, C, D (2) B, C, A, D

(3) D, C, A, B (4) C, D, B, A

41. In a _____ search, we begin search at a starting node, and when we come across a dead end then we backtrack.

(1) Breadth first (2) Connected

(3) Depth first (4) Bubble

42. The length or the number of data elements of the array can be obtained from the index set by which of the following formula?

(1) Lower Bound – Upperbound – 1

(2) Upperbound – Lowerbound + 1

(3) Lowerbound – Upperbound + 1

(4) Upperbound + Lowerbound – 1

43. Which of the following is the correct sequence of books from top to bottom books based on the given statements?

Consider the following statement:

A. There are seven books, one each on Psychology, Hindi, English, Sociology, Economics, Education and Accountancy lying on a table one above the other.

B. Sociology is on top of all books

C. Accountancy is immediately below education which is immediately below Sociology

D. Economics is immediately above Psychology but not in the middle

E. Hindi is immediately below Psychology.

Choose the correct answer from the options given below:

(1) Hindi, Psychology, Sociology, English, Education Accountancy, English

(2) Sociology, Education, Accountancy, English, Economics, Psychology, Hindi

(3) English, Economics, Education, Hindi, Psychology, Sociology, Accountancy

(4) Sociology, English, Accountancy, Education, Economics, Psychology, Hindi

44. Choose the odd one from the following :

(1) Crow (2) Pigeon

(3) Parrot (4) Butterfly

45. Which of the following is a Universal logic gate ?

(1) AND (2) OR

(3) NAND (4) XOR

46. The area of the region $\{(x, y): x^2 \leq y \leq x\}$ is given by:

(1) $\frac{1}{6}$ sq. units (2) $\frac{1}{3}$ sq. units

(3) $\frac{1}{2}$ sq. units (4) $\frac{2}{3}$ sq. units

47. Identify the letters which have their mirror images identical to the letter itself are:

A. J B. K

C. A D. H

Choose the correct answer from the options given below :

(1) C and D only (2) A and D only

(3) A and C only (4) B and D only

48. Which state occurs when two or more processes are waiting indefinitely for an event that can be caused only by one of the waiting processes.

(1) Deadlock state

(2) Synchronisation state

(3) Resource allocation state

(4) Preemption state

49. In the following question, various terms of an alphabet series are given with one term missing as shown by (?). Choose the missing terms out of the given alternatives.

M, N, O, L, R, I, V, ?



- (1) A (2) E
(3) F (4) H

50. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Deadlocks occur in a multi programming environment when each thread in a set is waiting for an event that can be caused only by another thread in the same set.

Reason (R): A thread must always request a resource before using it and release it after using it. If requested resources are not available the thread enters a waiting state until the resources becomes free.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct

51. The fundamental function of a register in digital circuits is to :

- (1) Perform Arithmetic operations
(2) Store Binary information
(3) Compare two numbers
(4) Encode input signals

52. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A):

flop flops are used in counters and registers.

Reason (R): flip flops are combinational circuits that store data temporarily.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

53. Which of the following relationship are correct?

A. Chocolate : Sugar :: Building : Cement

B. Sound : Feeble :: Colour : Faded

C. Dubious : Indisputable ::

Painful : Tormenting

D. Calligraphy : Writing :: Chapter : Stanza

E. Restaurant : Menu :: Library : Catalogue

Choose the correct answer from the options given below :

- (1) A, B and C (2) B, C, D only
(3) C and D only (4) A, B and E only

54. In a _____ linked list, the last node's link field points to the first node of the list.

- (1) Circular (2) Linear
(3) Sequential (4) Indexed

55. Sequence the following to compute the effective access time needed to service a page fault.

A. Receive an interrupt from the disk I/O subsystem

B. Trap the operating system and save the user registers and process state

C. Determine that the interrupt was a page fault

D. Correct the page table and other tables to show that the desired page is now in memory

E. Restore the user registers, process state and new page table, and then resume the interrupted instruction

Choose the correct answer from the options given below :



- (1) A, B, C, D and E (2) B, C, D, E and A
(3) A, D, B, C and E (4) B, C, A, D and E

56. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): When lightning strikes, the sound is heard a little after the flash is seen.

Reason (R) : The Velocity of light is greater than that of the sound.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct

57. Match List - I with List - II.

	List - I		List-II
A.	$(\sqrt{2} + 1) + 1 + (\sqrt{2} - 1) + \dots \infty$	I.	$\frac{19}{24}$
B.	$\frac{1}{2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^4} + \frac{1}{2^5} + \frac{1}{3^6} + \dots \infty$	II.	6
C.	$6^{\frac{1}{2}} \times 6^{\frac{1}{4}} \times 6^{\frac{1}{8}} \dots \infty$	III.	$8(2 + \sqrt{2})$
D.	$8 + 4\sqrt{2} + 4 + \dots \infty$	IV.	$\frac{4 + 3\sqrt{2}}{2}$

Choose the correct answer from the options given below :

- (1) A-I, B-II, C-III, D-IV
(2) A-IV, B-I, C-II, D-III

- (3) A-IV, B-I, C-III, D-II
(4) A-I, B-IV, C-III, D-II

58. Match List - I with List - II.

	List - I (Registers)		List - II (Description)
A.	Context Register	I.	Read by the Host to get input
B.	Status Register	II.	Written by the host to sent output
C.	Data - in Register	III.	Contain bits that can be read by the host
D.	Data - out Register	IV.	Can be written by the host to start a command

Choose the correct answer from the options given below :

- (1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-I, B-II, C-IV, D-III

59. Arrange the following determinants in decreasing order.

A.	$\begin{vmatrix} 1 & 3 & 5 \\ 2 & 6 & 10 \\ 31 & 11 & 38 \end{vmatrix}$	B.	$\begin{vmatrix} 67 & 19 & 21 \\ 39 & 13 & 14 \\ 81 & 24 & 26 \end{vmatrix}$
C.	$\begin{vmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 3 & 5 & 2 \end{vmatrix}$	D.	$\begin{vmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{vmatrix}$

Choose the correct answer from the options given below :

- (1) A, B, C, D (2) B, A, D, C
(3) C, D, A, B (4) C, A, D, B

60. Arrange the following steps for retrieving data from the memory unit.

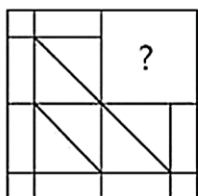
- A. The Memory Address Register (MAR) sends the address to memory chips
B. The address is placed on the address bus
C. The control unit sends a signal
D. The data is retrieved and placed in to the Memory Data Register (MDR)



Choose the correct answer from the options given below :

- (1) D, C, B, A (2) A, B, C, D
(3) A, C, D, B (4) B, D, A, C

61. Select a figure from amongst the four alternatives, which when placed in the blank space of fig (?) would complete the pattern.



62. Let the mean and variance of 12 observation be $\frac{9}{2}$ and 4, respectively. Later on, it was observed that two observations were considered as 9 and 10 instead of 7 and 14 respectively. If the correct variance is $\frac{m}{n}$, where m and n are coprime, then m + n is equal to:

- (1) 316 (2) 315
(3) 314 (4) 317

63. Which of the following is NOT an application of stack?

- (1) Towers of Hanoi
(2) Evaluation of Postfix expression
(3) Recursion
(4) Priority Queue

64. Which of the following are Combinational Circuit?

- A. Half adder
B. SR flip-flop
C. Full adder
D. D flip-flop

Choose the correct answer from the options given below:

- (1) A and C only
(2) B and D only
(3) B, C, D only
(4) A, B, C, D

65. Match List - I with List - II.

	List - I (Description of the Event)		List - II (Set theoretic Notation)
A.	Neither A nor B	I.	$(A \cap \bar{B}) \cup (\bar{A} \cap B)$
B.	At least one of A, B or C	II.	$\bar{A} \cap \bar{B}$
C.	Exactly one of A and B	III.	$A \cup B \cup C$
D.	All three of A, B and C	IV.	$A \cap B \cap C$

Choose the correct answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-II, B-IV, C-I, D-III

66. Match List - I with List - II.

	List - I		List - II.
A.	AND	I.	output 1 only if both inputs are 1
B.	OR	II.	output 1 if any input is 1
C.	NAND	III.	output 0 only if both inputs are 1
D.	NOR	IV.	output 1 only if both inputs are 0

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I (2) A-I, B-II, C-III, D-IV
(3) A-I, B-III, C-IV, D-II (4) A-II, B-I, C-IV, D-III

67. The target thread periodically checks whether it should terminate, allowing it an opportunity to terminate itself in an orderly fashion. This is known as:



- (1) Synchronous Cancellation
- (2) Asynchronous Cancellation
- (3) Default Cancellation
- (4) Deferred Cancellation

68. Choose the sets of numbers from the four alternative sets, that is similar to the given set:

Given set:

(3, 18, 36)

A. (5, 25, 50)

B. (2, 10, 16)

C. (4, 24, 48)

D. (5, 30, 60)

Choose the correct answer from the options given below:

(1) A and B only

(2) B and C only

(3) C and D only

(4) C and A only

69. Match List - I with List - II.

	List - I		List - II.
A.	$\frac{d}{dx}(\cos^{-1}x)$	I.	$\frac{1}{1+x^2}$
B.	$\frac{d}{dx}(\cot^{-1}x)$	II.	$\frac{1}{ x \sqrt{x^2-1}}$
C.	$\frac{d}{dx}(\operatorname{cosec}^{-1}x)$	III.	$-\frac{1}{\sqrt{1-x^2}}$
D.	$\frac{d}{dx}(\sec^{-1}x)$	IV.	$-\frac{1}{ x \sqrt{x^2-1}}$

Choose the correct answer from the options given below:

(1) A-I, B-II, C-III, D-IV (2) A-II, B-III, C-IV, D-I

(3) A-III, B-IV, C-I, D-II (4) A-III, B-I, C-IV, D-II

70. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Let f be a function defined by $f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

Assertion (A): $x = 0$ is point of minima of f .

Reason (R): $f'(0) = 0$

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

71. The value of $\int_0^{\pi/2} \frac{\sin^2 x}{1 + \sin x \cos x} dx$ is:

(1) $\frac{\pi}{3\sqrt{3}}$

(2) $\frac{\pi}{2\sqrt{2}}$

(3) $\frac{\pi}{9}$

(4) $\frac{\pi}{\sqrt{3}}$

72. If the foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, then the equation of the hyperbola having its eccentricity 2 is:

(1) $\frac{x^2}{4} - \frac{y^2}{12} = 1$

(2) $\frac{x^2}{16} - \frac{y^2}{12} = 1$

(3) $\frac{x^2}{12} - \frac{y^2}{4} = 1$

(4) $\frac{x^2}{2} - \frac{y^2}{9} = 1$

73. Arrange the following parabolas in increasing order of the length of their latus - rectum.

A. $y^2 = 8x$

B. $4x^2 + y = 0$

C. $y^2 - 4y - 3x + 10 = 0$

D. $y^2 - 4y + 4x = 0$

Choose the correct answer from the options given below:

(1) A, B, C, D

(2) B, C, A, D

(3) C, B, A, D

(4) B, C, D, A

74. Which of the following are characteristics of a Combinational Circuit?

A. Output depends only on present input

B. Output depends only on past input

C. It has no memory element

D. It uses flip flops to store data

Choose the correct answer from the options given below:

(1) A and B only

(2) A and C only

(3) B and D only

(4) C and D only

75. Let the equation of the pair of lines, $y = px$ and $y = qx$ can be written as $(y - px)(y - qx) = 0$. Then, the equation of the pair of the angle bisectors of the lines $x^2 - 4xy - 5y^2 = 0$



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Answer key & Solution



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(1) $x^2 - 3xy + y^2 = 0$

(2) $x^2 + 4xy - y^2 = 0$

(3) $x^2 + 3xy - y^2 = 0$

(4) $x^2 - 3xy - y^2 = 0$

ANSWER KEY

1.	2.	3.	4.	5.
c	d	c	b	a
6.	7.	8.	9.	10.
a	b	c	a	b
11.	12.	13.	14.	15.
b	c	b	b	a
16.	17.	18.	19.	20.
b	d	a	d	b
21.	22.	23.	24.	25.
a	c	d	d	c
26.	27.	28.	29.	30.
c	d	b	a	d
31.	32.	33.	34.	35.
a	b	b	b	c
36.	37.	38.	39.	40.
d	b	a	c	b
41.	42.	43.	44.	45.
c	b	b	d	c
46.	47.	48.	49.	50.
a	a	a	b	b
51.	52.	53.	54.	55.
b	c	d	a	d
56.	57.	58.	59.	60.
a	b	b	c	b
61.	62.	63.	64.	65.
b	d	d	a	b
66.	67.	68.	69.	70.
b	d	c	d	b
71.	72.	73.	74.	75.
a	a	d	b	c

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